

Lab 12: Projectile 2 (Non-Symmetric)

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Class: **PHYS 2125 (15921)**

Date: **2025-04-11**

Objective

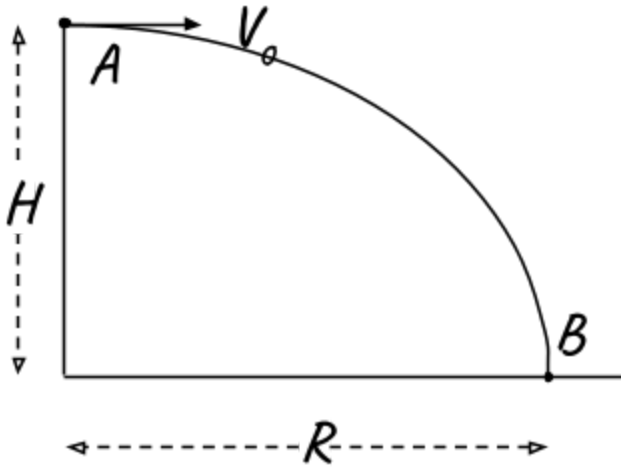
To determine the speed of the projectile in two different methods and compare the values.

Equipment

This experiment was simulated using http://phet.colorado.edu/sims/html/projectile-motion/latest/projectile-motion_all.html.

Theory

Part A - Horizontal Projectile



$$\uparrow v_{0y} = 0; a = -g; \Delta y = -H; t = t$$

$$\rightarrow v_{0x} = v_0; a = 0; \Delta x = R; t = t$$

$$\rightarrow \Delta x = v_{0x}t + \frac{1}{2}at^2$$

$$R = v_0t + 0 \text{ (Equation 1)}$$

$$\uparrow \Delta y = v_{0y}t + \frac{1}{2}at^2$$

$$-H = 0 + \frac{1}{2}(-g)t^2$$

$$t = \sqrt{\frac{2H}{g}} \text{ (Equation 2)}$$

$$R = v_0\sqrt{\frac{2H}{g}}$$

$$v_0 = \frac{R}{\sqrt{\frac{2H}{g}}}$$

where

v_0 is the initial velocity (horizontal, positive),

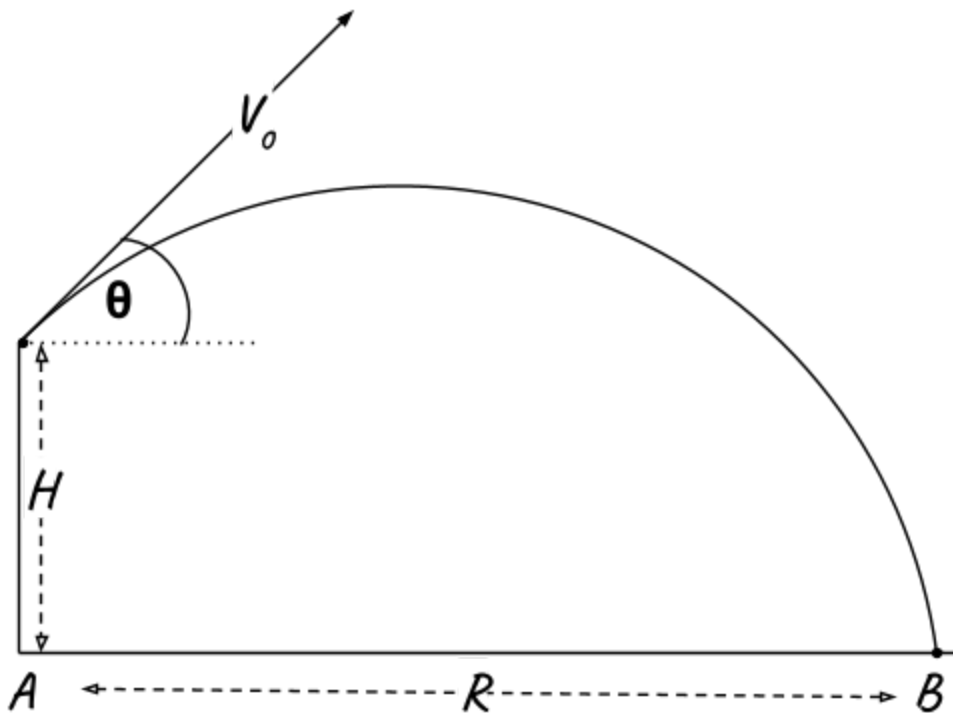
R is the range,

H is the starting (maximum) height,

A is the starting point, and

B is the landing point.

Part B - Projecting From a Height



In the horizontal x direction

$$\rightarrow v_{0x} = v_0 \cos \theta; a = 0, t = T, \Delta x = R.$$

And in the vertical y direction

$$\uparrow v_{0y} = v_0 \sin \theta; a = -g, t = T, \Delta y = 0.$$

$$\rightarrow \Delta x = v_{0x}t + \frac{1}{2}at^2$$

$$R = (v_0 \cos \theta)t + \frac{1}{2}(0)t^2,$$

$$t = \frac{R}{v_0 \cos \theta},$$

$$\uparrow \Delta y = v_{0y}t + \frac{1}{2}at^2,$$

$$-H = (v_0 \sin \theta)t + \frac{1}{2}(-g)t^2,$$

$$-H = (v_0 \sin \theta)\left(\frac{R}{v_0 \cos \theta}\right) - \frac{1}{2}g\left(\frac{R}{v_0 \cos \theta}\right)^2,$$

$$-H = R \tan \theta - \frac{1}{2}g \frac{R^2}{v_0^2 \cos^2 \theta},$$

$$\frac{1}{2}g\frac{R^2}{v_0^2\cos^2\theta} = R\tan\theta + H,$$

$$\Rightarrow v_0^2\cos^2\theta = \frac{gR^2}{2(R\tan\theta + H)},$$

$$v_0^2 = \frac{gR^2}{2\cos^2\theta(R\tan\theta + H)},$$

$$v_0 = \sqrt{\frac{gR^2}{2\cos^2\theta(R\tan\theta + H)}}.$$

where

v_0 is the initial velocity,

R is the range,

H is the initial height,

A is the starting point, and

B is the landing point.

Procedure

Part A

The following procedure was followed.

Initial Setup

1. Selected the "lab" option.
2. Left all settings at their default value.
3. Set the angle to 0° (towards the positive x-axis).

Trial (completed twice for each height H)

1. The height of the cannon was adjusted.
2. The distance R was measured.
3. v_0 was calculated.

Part B

The following procedure was followed.

Initial Setup

1. Selected the "lab" option.
2. Left all settings at their default value.
3. Set the height to 10 meters.

Trial (completed twice for each angle θ)

1. The angle θ of the cannon was adjusted.
2. The distance R was measured.
3. v_0 was calculated.

Data

Part A

	⬆ ⬇ ⬆ R ⬆ ⬇ ⬆ (m)	$v_0(s)$ ⬆ ⬇ ⬆
H ⬆ ⬇ ⬆ (m)	⬆ ⬇ ⬆	⬆ ⬇ ⬆
6	19.91	17.99259
7	21.5	17.988191
8	22.99	17.992521
9	24.38	17.989144
10	25.7	17.99
11	26.96	17.993746
Average		17.991032
StdDev		0.00203

Part B

	H (m)	R (m)	$\cos\theta$ (m)	$v_0(s)$
$\theta(\text{deg})$				
25	10.0	39.15	0.906308	17.997851
30	10.0	40.76	0.866025	18.000649
35	10.0	41.67	0.819152	17.999453
40	10.0	41.8	0.766044	18.000187
45	10.0	41.07	0.707107	18.000156
50	10.0	39.44	0.642788	17.998693
Average				17.999498
StdDev				0.000966

Results

The average value of v_0 between the two parts was 17.9953.

$$v_{0_a} = 17.9910 \pm 0.0020 \quad v_{0_b} = 17.9995 \pm 0.0010$$

These two values differ by 0.0005%.